A pad abort test is a kind of test of a launch escape system which conducted by setting the system along with the spacecraft still on the ground and let the system activate to carry the spacecraft flying away, then separate in the air and make the spacecraft land safely. The purpose of the test is to determine how well the system could get the crew of a spacecraft to safety in an emergency on the launch pad. As the spacecraft is set still on the ground, the test is also called "zero-altitude abort test" in against "high-altitude abort test".

Project Mercury

Section sources.

The Mercury program included several pad abort tests for the launch escape system with a boilerplate crew module.

1959 July 22 – First successful pad abort flight test with a functional escape tower attached to a Mercury boilerplate

1959 July 28 – A Mercury boilerplate with instruments to measure sound pressure levels and vibrations from the Little Joe test rocket and Grand Central abort rocket/escape tower

Project Apollo

The Apollo program included several pad abort tests for the launch escape system with a boilerplate crew module.

Pad Abort Test-1 was conducted on November 7, 1963, and

Pad Abort Test-2 was conducted on June 29, 1965.

Both tests were conducted at the White Sands Missile Range.

Shenzhou

China Manned Space Program included one pad abort test for Shenzhou spacecraft, called "zero-altitude flight test"(Chinese: 零高度飞行试验), conducted on October 19, 1998 9:00 am, one year prior to first Shenzhou spacecraft mission with a successful result.

Orion

The Orion pad abort test was conducted on May 6, 2010 at the White Sands Missile Range in New Mexico. The Launch Abort System lifted the Orion Boilerplate to a height of approximately 6000 feet above the ground and landed 6,900 feet downrange about 150 seconds later. The Abort test resulted in no damage to the test article and the mission was considered a complete success.

Commercial Crew

Dragon 2

The SpaceX Dragon 2 Pad Abort Test was conducted on May 6, 2015 at approximately 0900 Eastern Daylight Time (EDT). (A video clip released by SpaceX shows the timestamp of the moment of launch as 13:00:00). The vehicle splashed down safely in the ocean to the east of the launchpad 99 seconds later. A fuel mixture ratio issue was detected after the flight in one of the eight SuperDraco engines, but did not materially affect the flight. More detailed test results were to be subsequently analyzed by both SpaceX and NASA engineers.

Starliner

The pad abort test of Boeing's Starliner craft was conducted at 14:15 UTC on November 4, 2019 at the White Sands Missile Range. The capsule was lifted to 1,350 m (4,430 ft) and landed with airbags approximately 90 seconds after liftoff. Though the test was deemed a success, one of three main parachutes failed to deploy properly.

ISRO pad abort test

On 5 July 2018, Indian Space Research Organisation successfully conducted a pad abort test at Satish Dhawan Space Centre, Sriharikota. A first in a series of tests to qualify a crew escape system.

After a smooth countdown of five hours, the crew escape system, along with the simulated crew module with a mass of 12.6 tonnes, lifted off at 07.00 AM (IST) at the opening of the launch window from its pad at Satish Dhawan Space Centre, Sriharikota. The test was over in 259 seconds, during which the crew escape system along with crew module soared skyward, then arced over the Bay of Bengal and floated back to Earth under its parachutes about 2.9 km from Sriharikota.

Mengzhou

China Manned Space Program included one pad abort test for the Mengzhou spacecraft, called "zero-altitude escape flight test" (Chinese: 零高度逃逸飞行试验), conducted on June 17, 2025 12:30 am BJT from Site 95A of Jiuquan Satellite Launch Center with a successful result.

See also

Soyuz T-10-1, a Soyuz mission which ended with the use of the launch escape system

== References ==